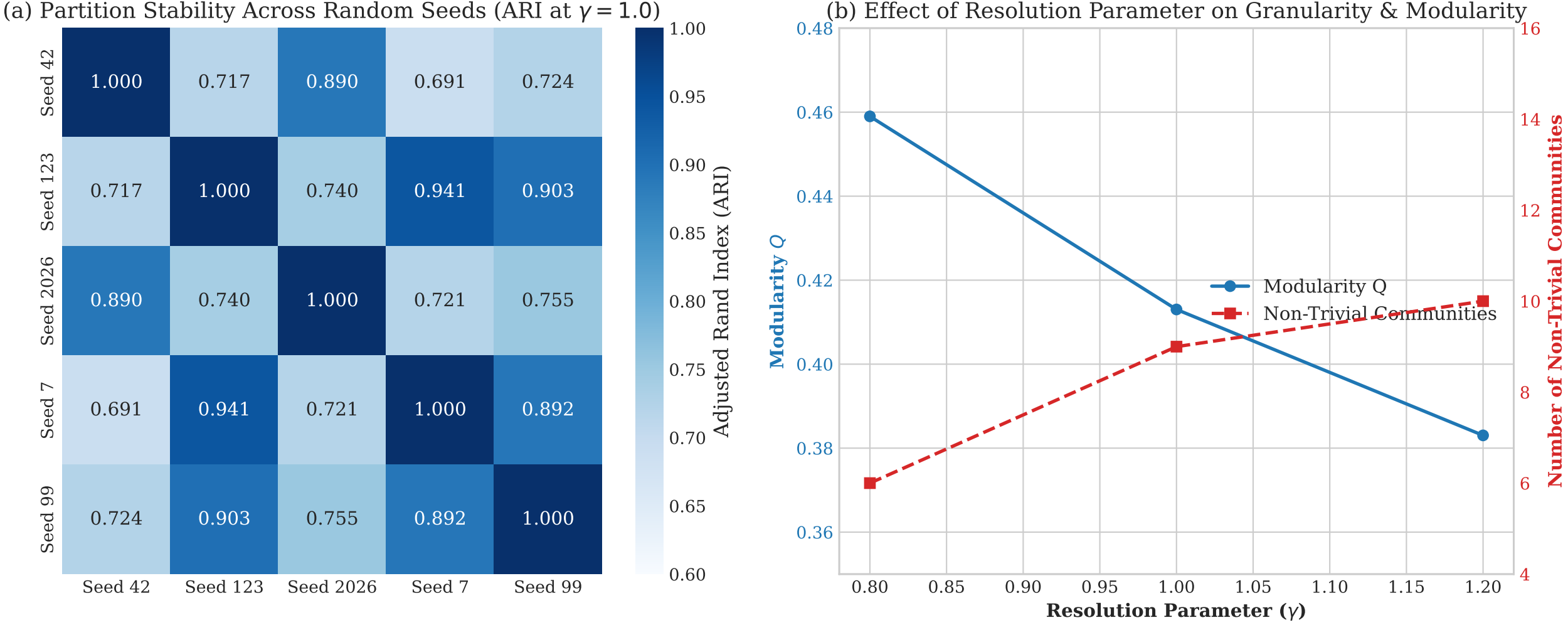
(a) Partition Stability Across Random Seeds (ARI at $\gamma = 1.0$)
Seed 42
Seed 123
Seed 2026
Seed 7
Seed 99
1.000 0.717 0.890 0.691 0.724
0.717 1.000 0.740 0.941 0.903
0.890 0.740 1.000 0.721 0.755
0.691 0.941 0.721 1.000 0.892
0.724 0.903 0.755 0.892 1.000
Seed 42
Seed 123
Seed 2026
Seed 7
Seed 99
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
Adjusted Rand Index (ARI)
(b) Effect of Resolution Parameter on Granularity & Modularity
0.48
0.46
0.44
0.42
0.40
0.38
0.36
Modularity Q
16
14
12
10
8
6
4
Number of Non-Trivial Communities
Modularity Q
Non-Trivial Communities
0.80
0.85
0.90
0.95
1.00
1.05
1.10
1.15
1.20
Resolution Parameter (γ)